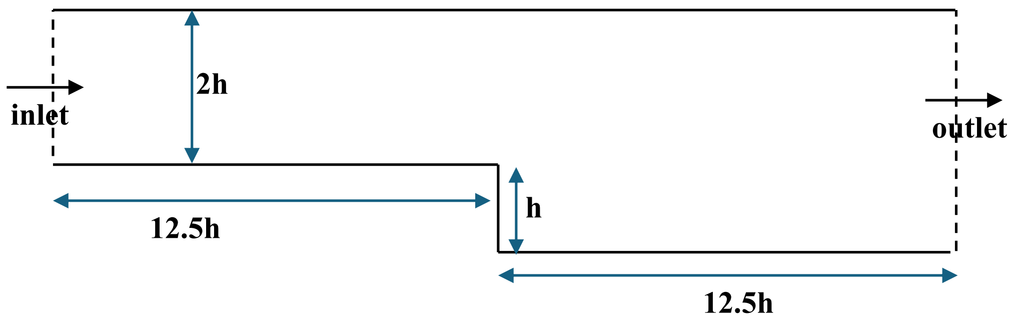
General goal: get familiar with the code: where does what?

Problem 6: Boundary conditions in a backward-facing step flow

A backward-facing step is a classic fluid mechanics problem involving flow over an abrupt, sudden increase in cross-sectional area. The flow behavior includes a separated shear layer, a recirculating eddy behind the step, a reattachment point, and a boundary layer recovery region. It is a common flow phenomenon in engineering applications, such as such as airfoils, vehicle wakes, combustors, and internal flows. The objective of this tutorial is to give you practical experience implementing and testing boundary conditions in the classic backward-facing step flow problem, investigating their influence on the recirculation zone, velocity and pressure field.

The flow 2D domain is sketched below. The dimensions are non-dimensional by considering the step height which is h=0.02m. Water is taken as the working fluid, density of 1000 kg/m^3, dynamic viscosity of 1e-3 Pa•s. For the inlet, $U_{max} = 0.25\ m/s$ is considered for a Reynolds number of $Re_h = \frac{U_{max}h}{\nu} = 5000$.



Boundary conditions to test

Case 1 – Benchmark

- **Inlet:** Uniform velocity $u = U_{max}, v = 0$
- **Outlet:** Zero-gradient for velocity, fixed pressure reference $p = 0$
- **Walls:** No-slip

This simulation allows you to establish a reference for Case 2, but also achieve proper simulation settings (grid resolution, timestep, iteration numbers, etc.) You can use the results (Figure 6-8) in the supplied paper (*3D-backward-steping-flow-Ansys.pdf*) to compare your results (Note the simulations in the paper are in 3D, whereas our current tests are in 2D).

Case 2 – Modified BCs

- **Inlet:** Fully developed parabolic profile $u(y) = U_{max}\left(1 - \left(\frac{2y}{y_{max}} - 1\right)^2\right)$
- **Outlet:** Zero-gradient outlet for pressure $\partial p/\partial x = 0$
- **Walls:** Slip wall on the top boundary $\partial u/\partial y = 0, v = 0$

You can choose the above BCs to change one at a time, or combined changes. Comparing the simulation results to the benchmark case to explore:

- How does the BCs affect the size and location of the recirculation zone?
- Does the reattachment length change?
- How does the pressure/velocity distribution differ?
- Does one outlet condition cause backflow or instability?
- Does converfence behavior change?